

Ex No: 1
11/12/2026

Verification of Kirchhoff's laws

Aim :-

To verify Kirchhoff's law of voltage and Kirchhoff's current law both theoretically and practically for a given DC circuit.

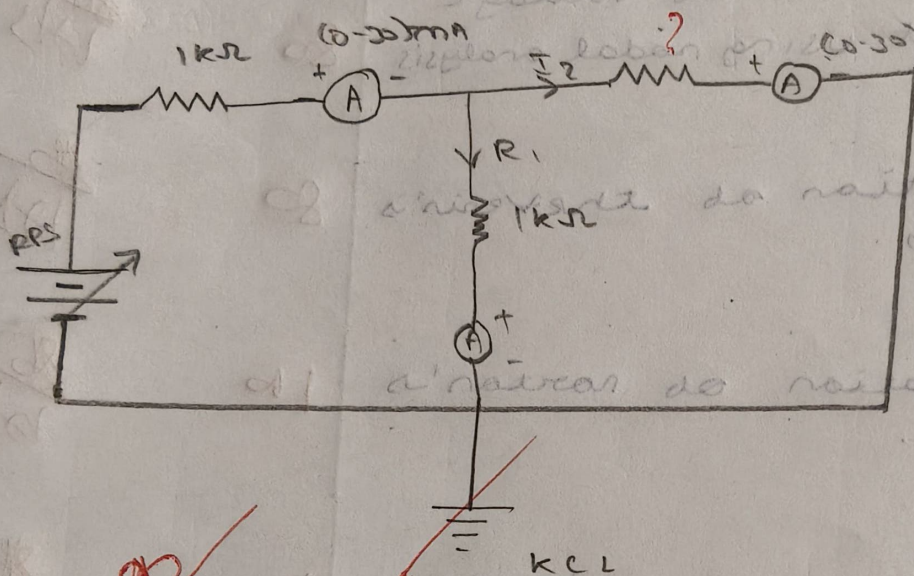
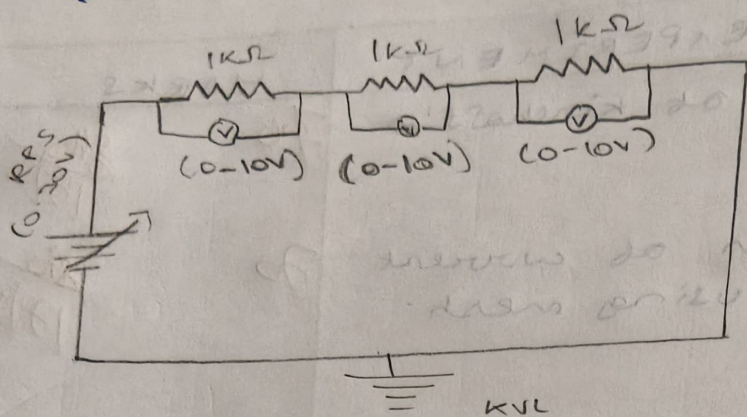
APPARATUS REQUIRED :-

Sl. No	APPARATUS	SPECIFICATION	QUANTITY
1)	Regulated Power Supply	(0-30V)	1
2)	Voltmeter	(0-30)V MC	3
3)	Ammeter	(0-10mA) MC	3
4)	Resistors	1k Ω	3
5)	Bread board	-	1

PROCEDURE :-

- 1) Give connection as per the circuit diagram
- 2) Switch on the supply vary the RPS and set a particular input voltage
- 3) Note down the readings of ammeter and voltmeter and tabulate them.
- 4) Vary the RPS to its minimum value and switch off the supply
- 5) Using the tabulated values, verify the

Circuit diagram :-



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V	V_1	V_2	V_3	$V = V_1 + V_2 + V_3$	I_1	I_2	I_3	$I = I_1 + I_2 + I_3$
2	0.61	0.61	0.61	1.83	0.57	0.58	0.57	1.72
6	1.97	1.97	1.97	5.91	1.67	1.68	1.67	5.02
9	2.89	2.89	2.89	8.68	2.49	2.52	2.49	7.50

9 verified

QUANTITY OF RESISTANCE

1	(0.30V)	(1) Regulated Power Supply
2	(0.30V)	(2) Voltmeter
3	(0.10mA)	(3) Ammeter
4	(1k)	(4) Resistor
5	(1k)	(5) Bread board

PROCEDURE

- 1) Draw the circuit diagram for the circuit diagram.
- 2) Switch on the supply and vary the RPS and note the voltmeter and ammeter readings.
- 3) Note down the readings of voltmeter and ammeter and calculate the power.
- 4) Vary the RPS to its minimum value and note the readings.
- 5) Vary the RPS to its maximum value and note the readings.

Marks - Split - UP ..

S.No	Description	Marks allotted	Marks obtained
1)	Experimental set-up connection	25 marks	20
2)	Observation	5 marks	20
3)	Calculation	5 marks	20
4)	Viva	5 marks	20
5)	Result	5 marks	10
Total			70

RESULT : ..

The ~~KIRCHHOFF'S~~ current law and voltage law are verified Practically and theoretically.

70
86
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